

Elliot Miller

734-223-7401 | [Website](#) | elliottgrantmiller@gmail.com | [linkedin.com/in/elliottgmiller](https://www.linkedin.com/in/elliottgmiller) | github.com/elliottmiller18

TECHNICAL SKILLS

Languages: C++ (17-23), Rust, C, Assembly (x86_64/ARM), Dafny, Python, C#

Libraries/Frameworks: SDL.h, Tokio, AWS CDK/SDK, Unity

Developer Tools: Git, AWS, AWS Lambdas, Docker, MacOS, Linux, NVim

HIGHLIGHTS

- Was ranked #3 on the Stack Overflow global leaderboards for discussing C++ language semantics (see my website for proof)

EXPERIENCE

Software Engineering Intern (Rust)

May 2026 – Aug 2026

Apple

Seattle, WA

- Developing new storage primitives from the kernel level for the Apple ecosystem and internal ML workloads

Software Engineering Intern (Rust)

Jan 2026 – May 2026

N1 (Founder's Fund)

New York, NY

- Contributed production code to a performant NASDAQ-like central limit order book matching engine handling more than \$2B in YTD volume and more than 100k trades per day at a startup backed by Founder's Fund, Kraken, Amber, and Anatoly Yakovenko
- $0 \rightarrow 1$ designed and implemented high-performance rate limiter inspecting every trader action (orders, deposits, etc.) between the NIC and matching engine layer to prevent DOS attacks
- Reduced all database CPU time by 98% (reducing usage from 4 constantly pinned cores to 1 mostly idle core) by indexing expensive queries and editing views to more efficiently join massive tables
- Live-mitigated critical issue halting all deposits and withdrawals

PROJECTS

GameBoy Emulator (C++23) [Github](#)

- Wrote a fully accurate Z80 Sharp CPU simulator/machine code interpreter capable of running GameBoy ROMs, emulating hardware bugs involving finicky instruction timing and interrupt handling
- Simulated all GameBoy-specific hardware components/systems, including internal clock circuits, interrupts, and proprietary graphics chip

Decentralized Peer to Peer File System (Rust) [Github](#)

- Wrote a decentralized P2P file-sharing system in Rust featuring a custom-built DHT based on the Kademlia protocol inspired by the design of BitTorrent.
- Guarded against Sybil and Eclipse attacks by implementing SKademlia security extensions including an upgrade to SHA-2 and POW requirements to join the network

Snake Compiler (Rust, x86)

- Wrote a x86 emitting, optimizing compiler for a Python/OCaml hybrid language called Snake
- Supports Rust interop, register allocation, dead code elimination, and dynamic typing, using lattices to eliminate unnecessary type checks

Multithreading Library (C++20)

- Wrote an efficient implementation of Mutexes and CVs similar to that of the C++ STL in a toy kernel that simulated a multi-core CPU for more realistic concurrent execution

EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Computer Science

Expected May 2027

- Relevant Coursework: Formal Verification of Systems Software, Distributed Systems, Computer Networks, Operating Systems, Compilers, Web Systems, Computer Organization, Data Structures and Algorithms, Computer Theory, Game Development, Programming Languages